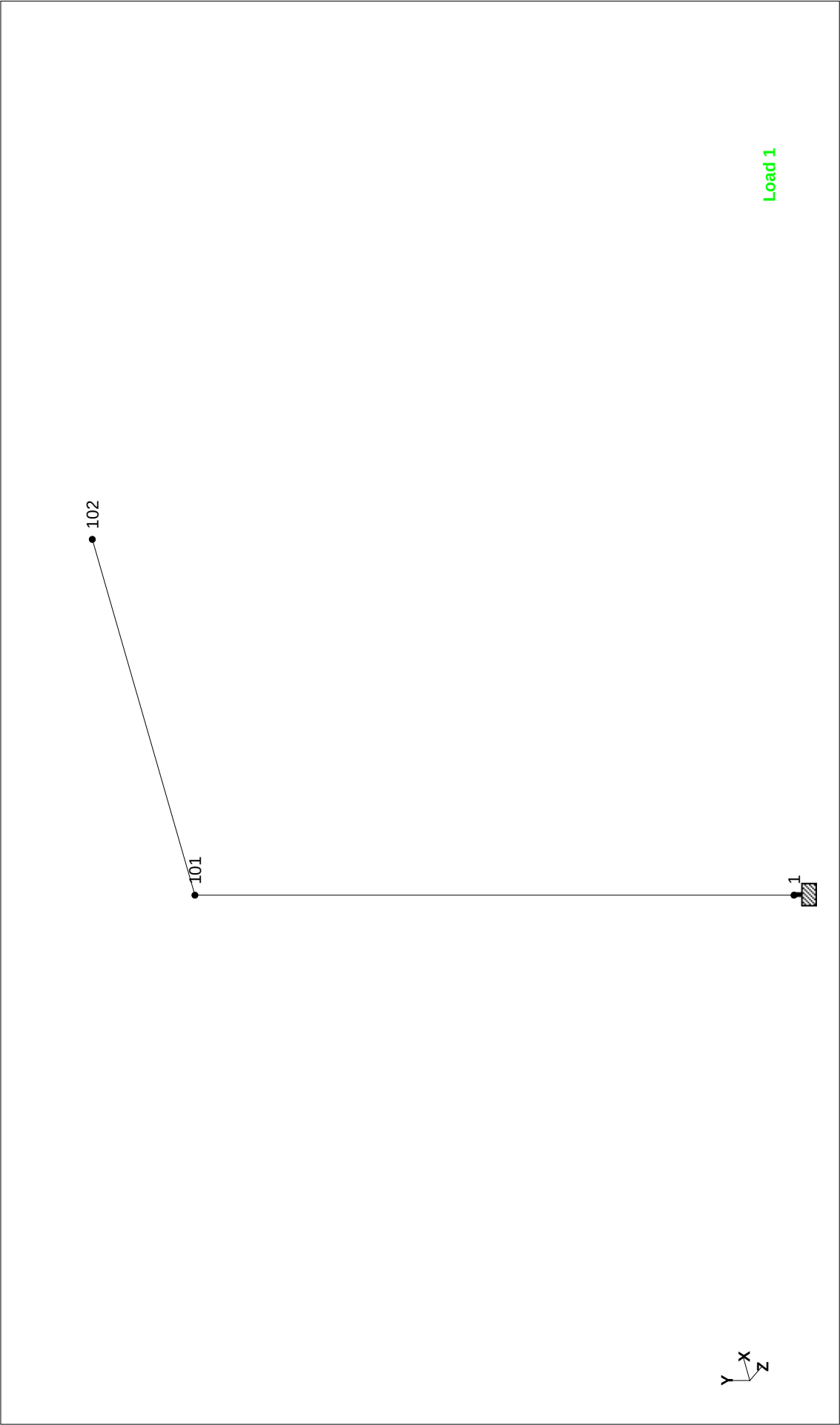
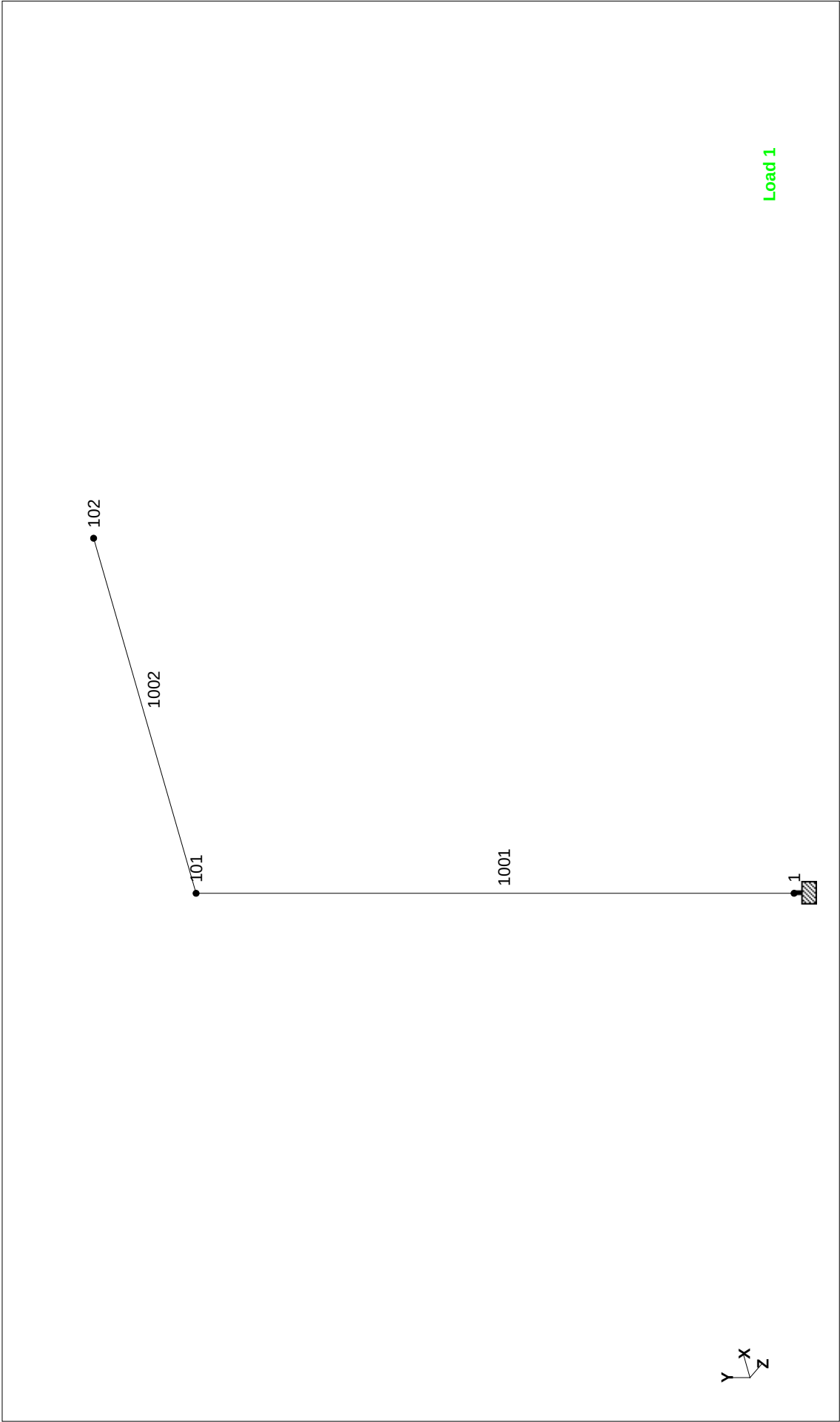


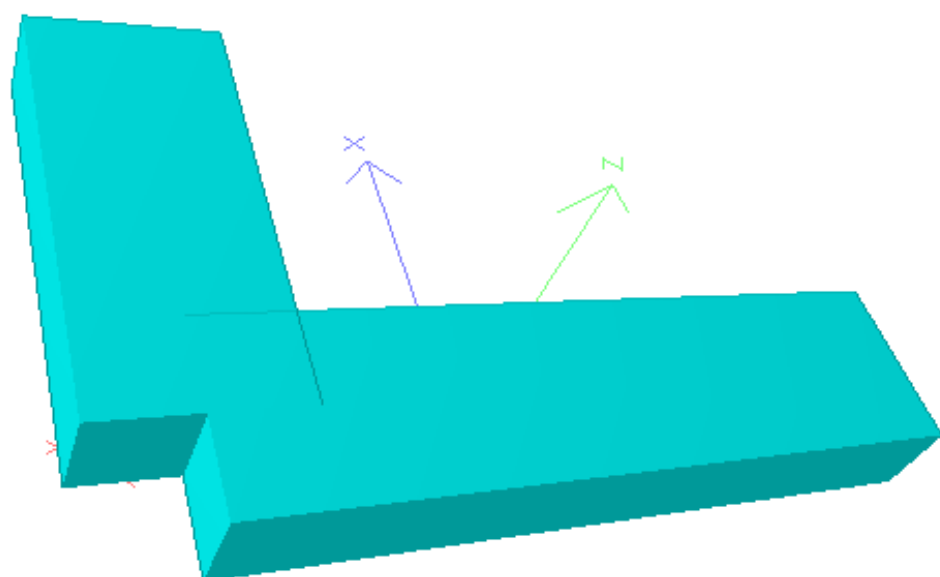
3.7 RC Design



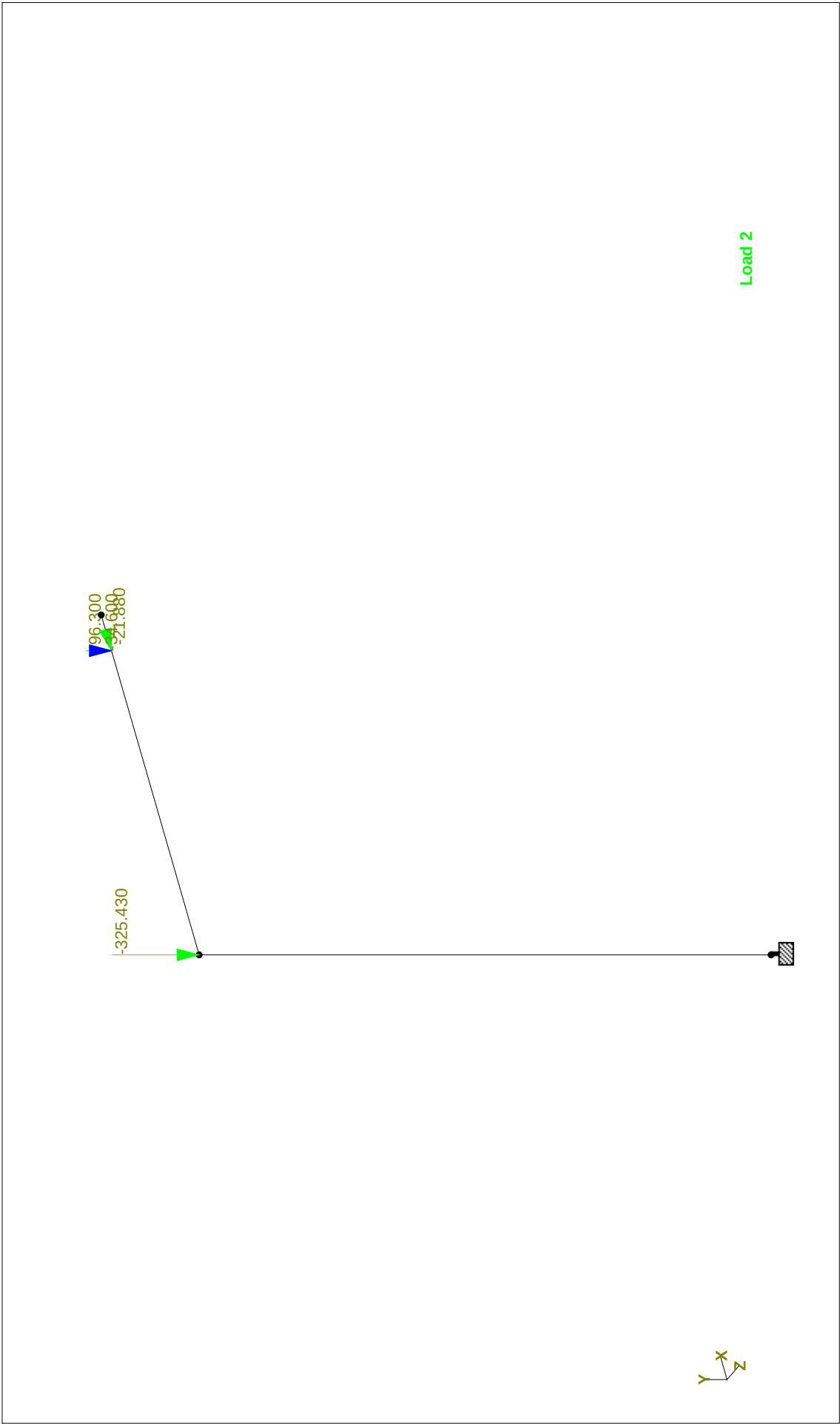
Load 1



MEMBER NO & NODE NO

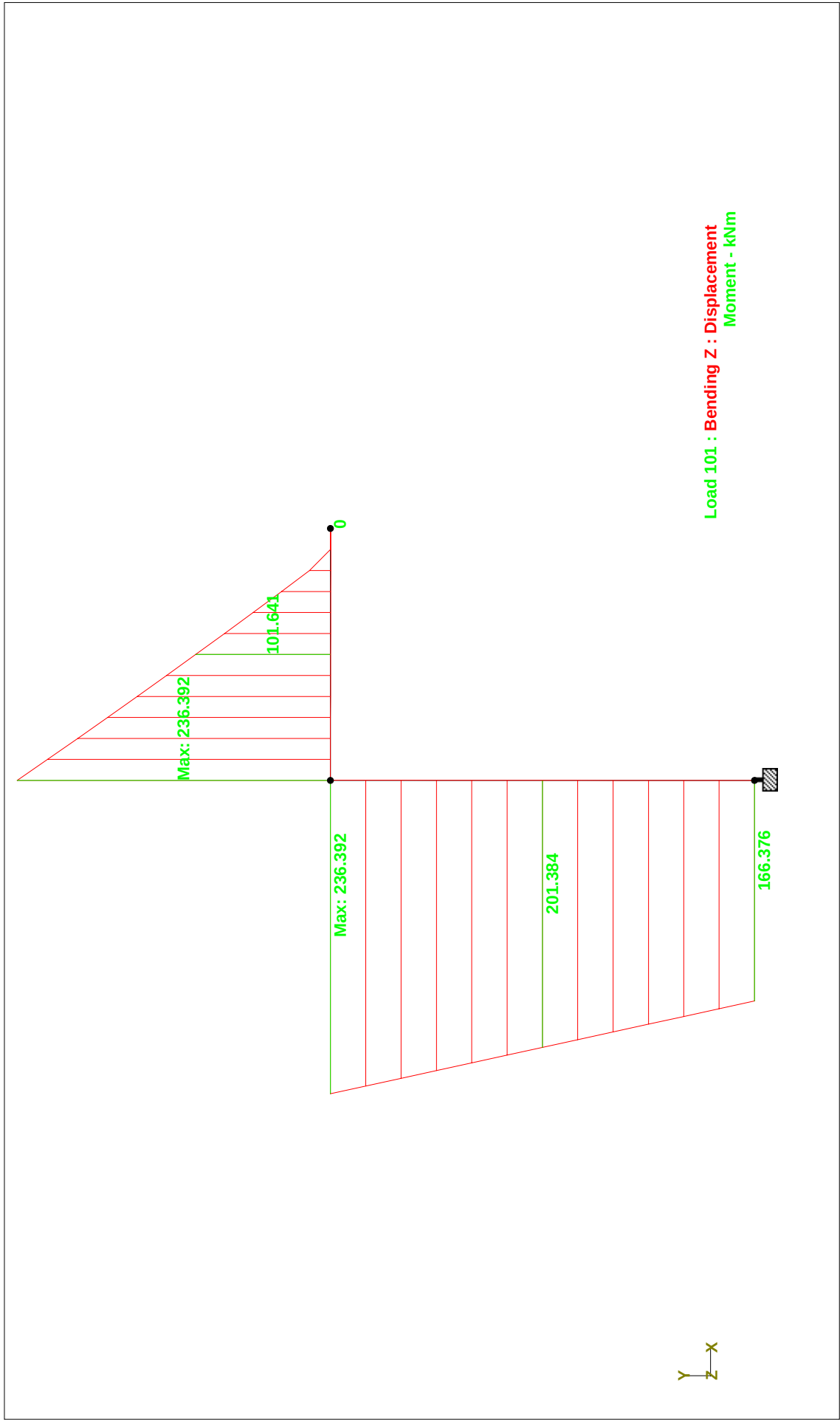


3D Rendered View

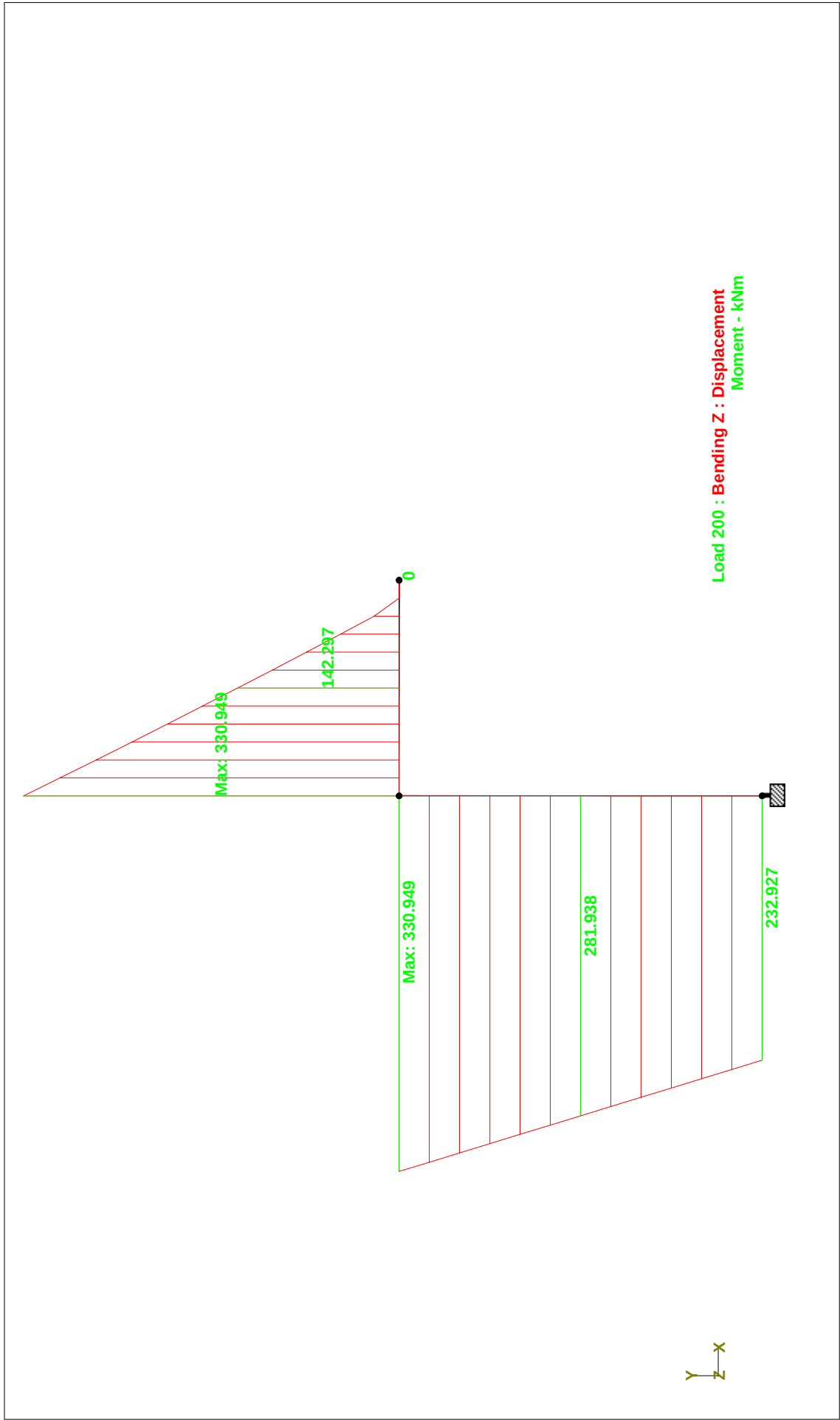


LOAD 2 DL

Load 2



BMD FOR FOOTING DESIGN



BMD FOR RCC DESIGN

0

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	104	0.81	82.34	18.49	0.00	0.00	0.00
105	0.78	38.04	5.80	0.00	0.00	0.00	0.00
2	104	0.00	93.84	21.17	0.00	0.00	0.00
105	-1.25	45.29	6.66	0.00	0.00	0.00	0.00
3	104	0.00	96.30	21.88	0.00	0.00	0.00
105	-0.07	46.12	6.89	0.00	0.00	0.00	0.00
4	104	0.00	96.14	21.87	0.00	0.00	0.00
105	-0.07	45.39	6.77	0.00	0.00	0.00	0.00
5	104	0.00	93.84	21.17	0.00	0.00	0.00
105	-1.23	45.14	6.64	0.00	0.00	0.00	0.00
6	104	-0.81	82.50	18.51	0.00	0.00	0.00
105	-0.87	38.93	5.95	0.00	0.00	0.00	0.00
101	104	0.04	-32.87	26.04	0.00	0.00	0.00
105	-1.01	11.17	-5.55	0.00	0.00	0.00	0.00
108	104	-0.03	-33.88	25.80	0.00	0.00	0.00
105	-1.93	14.71	-7.38	0.00	0.00	0.00	0.00
115	104	0.00	-36.94	28.12	0.00	0.00	0.00
105	-0.01	13.45	-6.42	0.00	0.00	0.00	0.00
122	104	0.00	-37.00	28.13	0.00	0.00	0.00
105	-0.01	13.76	-6.77	0.00	0.00	0.00	0.00
129	104	-0.01	-33.89	25.81	0.00	0.00	0.00
105	-1.86	14.60	-7.29	0.00	0.00	0.00	0.00
136	104	-0.01	-32.79	26.02	0.00	0.00	0.00
105	0.02	10.98	-5.30	0.00	0.00	0.00	0.00

Reaction used for RCB7 & Pedestal Design

***** END OF LATEST ANALYSIS RESULT *****

143. LOAD LIST 202 TO 207

144. PRINT MEMBER FORCES

0 MEMBER FORCES 0

```

*****
*
*          STAAD.Pro
*          Version  2006      Bld 1002.US
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=    FEB 12, 2009
*          Time=    14:59:31
*
*          USER ID: roark
*****

```

```

1. STAAD PLANE DUBAI METRO BLDG-03
INPUT FILE: AQD BLDG-03 (BRACKET +RCC PEDESAL )12 FEB 09.STD
2. START JOB INFORMATION
3. ENGINEER DATE 12-FEB-09
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 0 -2.2 -1.00044E-010; 101 0 1 -1.00044E-010; 102 1.9 1 -1.00044E-010
9. *****
10. MEMBER INCIDENCES
11. 1001 1 101; 1002 102 101
12. *****
13. MEMBER PROPERTY BRITISH
14. *****COL*****
15. 1001 PRIS YD 0.8 ZD 0.4
16. *****BEAM*****
17. 1002 PRIS YD 0.8 ZD 0.4
18. DEFINE MATERIAL START
19. ISOTROPIC CONCRETE
20. E 2.17185E+007
21. POISSON 0.17
22. DENSITY 24
23. ALPHA 1E-005
24. DAMP 0.05
25. END DEFINE MATERIAL
26. CONSTANTS
27. MATERIAL CONCRETE ALL
28. *****
29. SUPPORTS
30. 1 FIXED
31. *****
32. LOAD 1 SELFWEIGHT
33. SELFWEIGHT Y -1
34. *****
35. LOAD 2 DL
36. ***MAX COLUMN REACTION AT GRID 18/G FOR SLS= 301.43+24(CRANE LOAD)=325.43 KN
37. ** ( REF ATTACHEMENT OF DM001-D-GAD-WGO-DR-DCC-545039 - A2 )
38. JOINT LOAD
39. 101 FY -325.43
40. MEMBER LOAD

```

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41. **MAX. SUPPORT REACTION OF STEEL COLUMN REF ATTACHEMENT OF REACTION )
42. 1002 CON GY -96.3 0.2
43. 1002 CON GX -21.88 0.2
44. ***REACTION DUE TO SELF WEIGHT OF 300X600 BEAM
45. **SELF WEIGHT = 0.3 X 0.6 X24 = 34.6 KN
46. 1002 CON GY -34.6 0.2
47. *****
48. **          SERVICEABILITY*          **
49. *****
50. LOAD COMB 101 1.0DL
51. 1 1.0 2 1.0
52. *****
53. *          CONCRETE DESIGN          **
54. *****
55. LOAD COMB 200 1.4DL
56. 1 1.4 2 1.4
57. *****DEFLECTION*****
58. LOAD COMB 301 SELF+DL
59. 1 1.0 2 1.0
60. *****
61. PERFORM ANALYSIS

```

□

P R O B L E M S T A T I S T I C S

```

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      3/      2/      1
ORIGINAL/FINAL BAND-WIDTH=      1/      1/      6 DOF
TOTAL PRIMARY LOAD CASES =      2, TOTAL DEGREES OF FREEDOM =      6
SIZE OF STIFFNESS MATRIX =      1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE   =      12.0/    4603.0 MB

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□

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62. LOAD LIST 101
63. PRINT SUPPORT REACTION

```

□ SUPPORT REACTION

□

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
-------	------	---------	---------	---------	-------	-------	-------

[]

1	101	21.88	495.50	0.00	0.00	0.00	166.38
---	-----	-------	--------	------	------	------	--------

Conservative footing Design using maximum reactions values for all columns at grid 13/G to 18/G as force-Y includes max. column reactions @ grid 3-8/3-G (ref INPUT data line 36 Pg 89

***** END OF LATEST ANALYSIS RESULT *****

64. LOAD LIST 200

65. START CONCRETE DESIGN

[] CONCRETE DESIGN[]

66. CODE BS8110

PROGRAM CODE REVISION V1.5_8110_97/1

67. UNIT MMS NEWTON

68. MINSEC 10 ALL

69. FC 40 ALL

70. FYMAIN 460 ALL

71. CLEAR 50 MEMB 1001

72. ELY 2 MEMB 1001

73. TRACK 2 ALL

74. DESIGN COLUMN 1001

=====

C O L U M N N O . 1001 D E S I G N R E S U L T S

FY - 460. FC -40. N/MM2 RECT SIZE - 400. X 800. MM,

AREA OF STEEL REQUIRED = 1280. SQ. MM.

Provided steel = 14T16 = 2800 SQ.MM which is greater than 1280 SQ.MM. Hence Safe.

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION

12 12 MM	0.424	200	EACH END
(PROVIDE EQUAL NUMBER OF BARS AT EACH FACE)			

BRACED /SHORT in z E.L.z= 3200 mm (3.8.1.3 & 5)			
BRACED /SLENDER in y E.L.y= 6400 mm (3.8.1.3 & 5)			
END MOMS. MZ1= 232 MZ2= 330 MY1= 0 MY2= 0			
SLENDERNESS MOMTS. KNM: MOMZ= 0 MOMY= 35			
DESIGN LOADS KN METER: MOM.= 404 AXIAL LOAD= 693			
DESIGNED CAP. KN METER: Minimum Reinforcement.			

*****END OF COLUMN DESIGN RESULTS*****

75. DESIGN BEAM 1002

=====

B E A M N O. 1002 D E S I G N R E S U L T S - FLEXURE

LEN - 1900. mm FY - 460. FC - 40. SIZE - 400. X 800. mm

LEVEL	HEIGHT	BAR INFO	FROM	TO	ANCHOR
	mm		mm	mm	STA END
1	760.	6- 16 MM	0.	1900.	YES YES

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL (+VE/-VE) (SQ. MM)	MOMENTS (+VE/-VE) (KN-METER)	LOAD (+VE/-VE)
0.	0.0/ 0.0	0.00/ 0.00	0/ 200
158.	0.0/ 416.0	0.00/ 0.13	0/ 200
317.	0.0/ 416.0	0.00/ 21.92	0/ 200
475.	0.0/ 416.0	0.00/ 51.61	0/ 200
633.	0.0/ 416.0	0.00/ 81.57	0/ 200
792.	0.0/ 416.0	0.00/ 111.80	0/ 200
950.	0.0/ 452.5	0.00/ 142.30	0/ 200
1108.	0.0/ 550.3	0.00/ 173.07	0/ 200
1267.	0.0/ 649.0	0.00/ 204.10	0/ 200
1425.	0.0/ 748.5	0.00/ 235.41	0/ 200
1583.	0.0/ 848.9	0.00/ 266.99	0/ 200
1742.	0.0/ 950.2	0.00/ 298.83	0/ 200
1900.	0.0/ 1052.3	0.00/ 330.95	0/ 200

Provided steel = 4T20 = 1256 SQ.MM which is greater than 1052.3 SQ.MM. Hence Safe.

B E A M N O. 1002 D E S I G N R E S U L T S - SHEAR

PROVIDE SHEAR LINKS AS FOLLOWS

FROM - TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1 158 mm	0.0 kN	200	10 mm	2	158 mm
158 END 2	203.7 kN	200	10 mm	6	348 mm

Provided steel = 10T@100 which is greater than 10T150. Hence Safe.

```
____ 102J_____ 1900.X 400.X 800_____ 101J_____  
| |=====| |  
| 6No16 H 760. 0.TO 1900 |  
| |  
| 2*10c/c158 6*10c/c348 |  
| |  
| |  
| |  
|_____|  
  
_____  
|oooooo | |oooooo | |oooooo | |oooooo | |oooooo | |oooooo | | | | |
| 6T16 | | 6T16 | | 6T16 | | 6T16 | | 6T16 | | 6T16 |  
| | | | | | | | | | | | | | | |  
| | | | | | | | | | | | | | | |  
| | | | | | | | | | | | | | | |  
| | | | | | | | | | | | | | | |  
| | | | | | | | | | | | | | | |  
|_____| |_____| |_____| |_____| |_____| |_____| |_____|
```

*****END OF BEAM DESIGN*****

76. END CONCRETE DESIGN

77. FINISH

***** END OF THE STAAD.Pro RUN *****

REVISED FOOTING DESIGN FOR LATH ON PIT AREA (Grid 3-G/3-13 to 3-18)

Allowable Soil Bearing Pressure = 150 kN/m² (Footing 1 m below formation level) { From Green line Depot Ground Report Doc no # 544051 pg 30 of 38 table no 7.9)
Water Table at +9.9 DMD Level = 3.1 m Depth below GL
Allowable Soil Bearing Pressure = 100 kN/m² (Assumes Ground Water to be at surface level) { From Green line Depot Ground Report Doc no # 544051 pg 30 of 38 table no 7.9 , Notes 6 }
Effective SBC (Gross at Variable depth) = 100+0.5 (depth of footingX20) =Varies (kN/M²)
Soil Density (Dry) = 20 kN/m³
Soil Density (submerged) = 10 kN/m³
Concrete Density = 24 kN/m³

Location on Drawing	Footing Mark (in RL)	Size of Footing (in RL)	Load Taken from RL (page no-232)		Footing Mark (in GL)	Rev. Footing Size in GL For additional forces from Col-C39a @ 3-G/1/13-18	Fdn Depth (M)	Revised Loads on Footings				Bearing Pressure (kN/m ²) (due to vertical rea)	Bearing Pressure (kN/m ²) - due to Moment Mz	Total Bearing Pressure (kN/m ²)	Allowable soil Bearing Pressure (kN/m ²)	Remarks
			STAAD Column Reaction (ULS) (kN)	Max Vertical Crane Load Combination (kN)				Fy & Mz Ref. Pg-99 (based on max. reactions @18/G)	Footing Weight (kN)	Dry Soil Weight (kN)	Total Load Fy					
1	2	3	4	5	6a	6b	7	Fy = 8 Mz = 9	10	11	8+10+11	13	14	13+14	16	17
13/G	F5	2	0.6	354	F10	3.6	2.50	495.50	233.28	453.60	1182.38	91.23	21.40	112.63	150.00	O.K
14/G	F5	2	0.6	380	F10	3.6	2.50	495.50	233.28	453.60	1182.38	91.23	21.40	112.63	150.00	O.K
15/G	F5	2	0.6	325	F10	3.6	2.50	495.50	233.28	453.60	1182.38	91.23	21.40	112.63	150.00	O.K
16/G	F5	2	0.6	365	F10	3.6	2.50	495.50	233.28	453.60	1182.38	91.23	21.40	112.63	150.00	O.K
17/G	F5	2	0.6	381	F10	3.6	2.50	495.50	233.28	453.60	1182.38	91.23	21.40	112.63	150.00	O.K
18/G	F5	2	0.6	422	F10	3.6	3.20	495.50	233.28	635.04	1363.82	105.23	21.40	126.63	132.00	O.K

Note: For RL Design Dat please refer Design Document # DM001-D-GAD-WGO-DR-DCC-545039-A2